Conditions needed for life

17

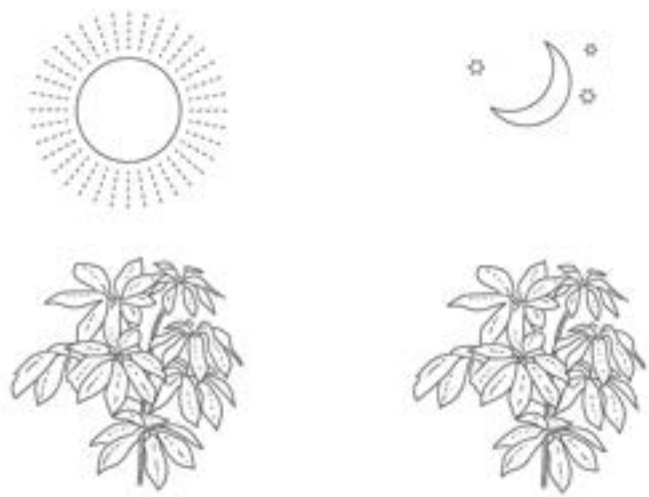
Life is a complicated process and many conditions are needed for life to continue.

1 Give five typical features of a living organism.

Different types of organisms require different conditions, but all need energy and raw materials to replace cells and build new ones. All living things have a *metabolism* (Gk. *metabol* change). Metabolism is the sum of the chemical reactions taking place in an organism. These reactions take in new materials and generate energy. Plants and animals have their own specific requirements, but both need energy and raw materials. Green plants are *autotrophs;* they can produce their own food by photosynthesis and generate energy through respiration.

Photosynthesis and respiration

Green plants use water and carbon dioxide (and enzymes) to produce their own food – known to us as carbohydrates. Light energy drives these reactions.



Photosynthesis and respiration

Respiration

Photosynthesis occurs when plants convert radiant energy into sugars, water and other forms of energy, which are then used by the plant for growth and to maintain its health. Plants contain specialised organelles called *chloroplasts* which are not found in animal cells. Only chloroplast-containing cells can photosynthesise. The chloroplast contains several pigments that trap light. The pigment chlorophyll absorbs the red and blue light rays that make up white light. There are several types of pigment in plants and four types of chlorophyll. When a plant is exposed to radiant energy (sunshine), the light rays cause a movement of electrons within the atoms of the leaf. This, in turn, activates the ATP/ADP. During this reaction energy is released that allows for the photosynthesis reaction to occur.

carbon dioxide + water + radiant energy + chlorophyll produce carbohydrates and oxygen.

Photosynthesis produces glucose as shown by the equation:

$$6H_2O + 6CO_2 + \text{energy} \rightarrow C_6H_{12}O_6 + 6O_2$$

All cells need a constant supply of energy and the energy is produced by respiration. Respiration involves many stages but can be summarised thus: $C_6H_{12}O_6 + 6O_2 \rightarrow 6H_2O + 6CO_2 + \text{energy}$.

Again, enzymes are an essential part of the reactions. Note: the energy above is equal to 2881.2 kJ.

2 Summarise, in words, the photosynthesis and respiration equations.

3 Discuss the statement: *'Photosynthesis only occurs when sunlight is shining on a green plant and respiration only occurs when a plant is in the dark.'*

4 Name another nutrient (not already mentioned) needed by plants.

Humans, like all animals, are *heterotrophs* – meaning they cannot manufacture their own food. Heterotrophs need complex organic compounds. The food humans eat must contain enough water, energy, vitamins, minerals and body-building substances to keep them healthy. Humans need a balanced diet, that is one that has all the substances we need and in all the right amounts.

Although water contains no energy and is not really a type of food, it is essential for our health. Water makes up approximately 70% of the body. Every day some water is lost through the kidneys, sweating and breathing out (exhaling). These losses must be replaced otherwise dehydration will occur. We could live for several weeks without food but only a few days without water. Some foods, especially vegetables, contain a large amount of water.

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5 Outline the major difference between an autotroph and a heterotroph.

6 Discuss the statement: '*Respiration is the reverse chemical reaction to photosynthesis*'.

7 What does this statement define: '*A process involving an unusual loss of body fluids*'?

8 What does this describe: '*Originally called vital amines, these naturally occurring substances are needed in very small amounts for the regulation of metabolic processes.*'?

9 Why is water not considered a food?

10 Name two ways in which the body loses water.

11 If water is not considered a food, why is it so important to our health?

Indicate whether statements 12 to 16 below are true or false.

12 Some animals drink very little; they gain most of their water requirement from food.

13 Chlorophyll is a green substance in plants needed for photosynthesis.

14 People eating a balanced diet need to take extra vitamins to keep healthy.

15 Glucose is a type of mineral.

16 Enzymes are biological catalysts.

Energy is measured in *joules (J)*, named in honour of British physicist James Joule (1818–1899). A joule is a very small unit of energy, so we tend to use *kilojoules (kJ)* instead (1000 J = 1 kJ). Energy is needed for all living processes, which is why our body demands a continuous supply of food. During sleep the tissues of the heart, lungs, kidneys, liver and other organs continue to work. Even the brain and muscles, although relatively inactive, use a quantity of energy. The energy we use depends on what we are doing, and can vary considerably from person to person, but the following table shows some typical figures in kilojoules per hour.

Activity	kJ used per hour
sleeping	65
light housework	600
jogging	400
squash	900
writing at a desk	160
road digging	1300
tennis	700

17 What is the total energy used by a student carrying out the following activities? (Show working.)
- **a** Jogs 15 minutes to the shops to buy a paper and jogs back.
- **b** Spends 1 hour tidying up room, followed by 1 hour 30 minutes doing homework.
- **c** Tired by all this, a nap of 30 minutes.

18 A person who leads an inactive life uses less energy than a very active person. If we take in less energy through food than we use up, we get thinner. What happens if we take in more energy than our body uses?

Growth also requires energy. Adolescence is one example of this. Adolescence usually happens between the ages of 11 to 18 years and frequently growth occurs in spurts rather than in a regular manner. Adolescents need as much (or more) energy from foods as adults do.

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Carbohydrates are energy foods. The most common are starches and sugars. Plants, and food obtained from plants, are often high in carbohydrates. If we add a solution of iodine to a food containing starch it turns to a blue-black colour. Foods high in starch include barley, rice, wheat and potatoes. Different foods have different energy values. Sugars include sucrose (table sugar) and glucose. Glucose is very quickly absorbed into the bloodstream, which is why it is often given to invalids or patients recovering in hospital. (Whenever you see a food or ingredient that ends in '-ose', then it is a type of sugar.) Some sugars (eg glucose) can be detected by heating with Benedict's reagent, which will go from blue to brick-red.

19 Name a food that is rich in carbohydrates.

20 Suggest how giving glucose to recovering patients might be of benefit to them.

Fats and *oils* are even more concentrated energy foods than carbohydrates. They take longer to digest than carbohydrates. Fats tend to come from animal sources – meat, butter, eggs etc – but can also be obtained from plants, eg olive oil, sunflower oil etc. We need a certain amount of fat in our body, to store energy and act as insulation. Fats and oils give a pink colour when heated with a chemical called Sudan III and also leave a greasy spot when placed on paper and allowed to dry.

21 Name another food group apart from those already mentioned.

22 What is one way the body makes use of the food group you gave in question 21?

23 You think that a food contains starch and/or glucose. How would you test for these?

24 What type of food is *lactose*?

25 What is the major difference between a fat and an oil?

26 Why do we need a supply of fat in our body?

27 After exercise, athletes might take a drink containing glucose rather than 'ordinary' sugar (sucrose). Why might they do this?

28 In a hospital, plants and flowers are often given to patients. Why might these plants and flowers be removed from the patients' bedsides at night?

29 Why does the body use energy even when we are asleep?

30 Why can adolescents require as much food as an adult?

31 Aside from keeping our weight down, what are three other ways in which the body benefits from regular exercise?

32 There are many diets that claim to make people lose weight fast by following a special regime. Some advocate eating mainly protein and very little carbohydrate. Discuss why you should be careful before undertaking these (or any other) diets.

33 Many people follow a vegetarian diet. Outline the cases for and against being vegetarian.

34 What is the difference between a vegetarian and a vegan?

Give definitions for the following terms. You may need to use a text book or dictionary.

35 adipose tissue

36 obesity

37 nutritionist

All diets need a certain amount of cholesterol, which is used in making cell walls. However, some foods are very high in cholesterol. High levels of cholesterol in humans can cause health problems.

38 Name two foods that contain large amounts of cholesterol.

39 Outline some health problems associated with high levels of blood cholesterol.

40 Identify three ways of preserving food.

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